

Canadian Donations Mapping Platform

CDMP

Project Proposal

Team NACSS

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SECTION 1

Executive Summary

Although Elections Canada publicly releases donation records for political parties, candidates, and electoral associations, the data is difficult for most users to work with in practice. Users must download numerous raw data files, review and standardize data from different time periods, and create their own visualizations from scratch. As a result, analyzing this information requires strong data skills and can take several weeks, even for a trained research assistant.

This creates challenges for groups that would benefit from analyzing political finance information. Journalists covering campaign finance stories cannot quickly investigate donation trends, political science students may struggle to research topics such as per-capita giving across provinces, and civic organizations interested in democratic transparency lack accessible tools to explore the data effectively.

The CDMP platform addresses these challenges by transforming the entire workflow into a process that can be completed in under one minute. The proposed product is an interactive, map-centered web application that allows users to select geographic levels of analysis, apply filters by time period and political party, and explore donation patterns through visualized geographic data. Users will also be able to click on regions to view summary statistics, party-level breakdowns, and trend information.

The platform is designed with both public-facing and research-focused features. Public users will be able to access aggregated and anonymized donation data without requiring an account, while authenticated university-affiliated researchers will have access to more detailed individual-level records through a secure login system with appropriate privacy protections.

SECTION 2

Meeting Attendance

On Tuesday, May 26, all members of our team, Calvin Liew, Shahveer Rasool, Shamrita Saravanakumar, Abdul Hameed Mohammed, and Niveetha Sivakaran, attended our first team meeting. During the meeting, we discussed responsibilities for drafting the proposal and set up daily standups for Mondays, Thursdays, and Saturdays, in addition to our regular Tuesday meetings.

Personas

Persona 1: Maya Chen

Public Data Explorer



Bio

Maya Chen is a 21-year-old undergraduate journalism student interested in elections, campaign finance, and democratic transparency. She is comfortable using websites, maps, filters, and basic charts, but has no experience with programming, GIS, or raw data cleaning. She wants CDMP to help her quickly explore political donation patterns by party, year, and geography without needing to download or process Elections Canada data herself.

Profile

Age	21
Education	Bachelor's student in Journalism
Occupation	Full-time student
Location	Toronto, Ontario
Tech literacy	Medium
Access level	Public, no login required
Data visibility	Aggregated and anonymized donation data only
Platform	Browser-based web app
Personality tags	Reader, Extrovert, Curious

Primary Task

Quickly find a political donation pattern that could support a story, article, or class assignment.

User Story

As a journalism student, I want to filter political donations by party, year, and geographic level so that I can quickly identify donation patterns for a story or class assignment.

Core Needs

- Quickly generate a meaningful donation map without needing technical setup.

- Filter donation data by party, year, and geographic level.
- Click on a region to view clear summary statistics and trends.
- Understand the map through simple labels, legends, and explanations.
- Avoid working directly with raw CSV files, postal code matching, or GIS tools.
- Trust that the data is presented accurately, with clear notes for missing, suppressed, or limited data.

Frustrations

- Elections Canada donation data is difficult to find, download, and understand.
- Raw CSV files require too much manual cleaning before they become useful.
- She does not know how to match postal codes to electoral districts.
- Existing data does not immediately show geographic patterns visually.
- She may not know whether donation totals are misleading without population context.
- She needs quick, clear results because her work is often deadline driven.

Quote

"I need to quickly see whether there is a donation pattern worth writing about, without spending hours cleaning data first."

Success Criteria

Maya succeeds if she can open CDMP, apply basic filters, view a meaningful map, click a region, and understand whether there is a pattern worth investigating.

Persona 2: Amira Hassan

Civic Transparency Communicator



Bio

Amira Hassan is a 34-year-old program coordinator at a civic-engagement nonprofit in Ottawa. She creates public education materials about elections, democracy, and political transparency. She wants to use CDMP to explain broad donation patterns through clear, privacy-safe maps that community members, volunteers, and students can easily understand.

Profile

Age	34
Education	Bachelor's degree in Political Science
Occupation	Program Coordinator at a civic-engagement nonprofit
Location	Ottawa, Ontario
Tech literacy	Medium
Access level	Public, no login required
Data visibility	Aggregated and anonymized donation data only
Platform	Browser-based web app
Personality tags	Educator, Transparent, Responsible

Primary Task

Use aggregated donation maps to explain political finance patterns to a non-technical public audience.

User Story

As a civic transparency communicator, I want to view privacy-safe aggregated donation maps so that I can explain political finance patterns to the public.

Core Needs

- View donation patterns in a simple, visual format.
- Compare donations by region, party, and year.
- Use privacy-safe aggregated data without exposing individual donors.
- Explain political finance trends to non-technical audiences.
- Access clear map legends, labels, and summary statistics.
- Trust that the platform presents data responsibly and avoids misleading interpretations.

Frustrations

- Raw donation data is too technical for public education.
- Spreadsheets and complex charts are hard to explain to non-technical audiences.
- She does not want to expose individual donor information.
- She needs clear legends, labels, and plain-language summaries.
- It can be difficult to compare regions fairly without context like population size.
- She worries that users may misread the map if the data limits are not explained clearly.

Quote

“I want to help people understand where political donations come from, but I need the data shown in a clear and privacy-safe way.”

Success Criteria

Amira succeeds if she can use CDMP to explain donation patterns to a non-technical audience without exposing individual donor records or doing manual data processing.

Persona 3: Daniel Roy

Authenticated Research Assistant



Bio

Daniel Roy is a 26-year-old political science graduate research assistant working with a university research group in Toronto. He studies the spatial distribution of political finance in Canada and needs access to detailed donation records for academic research. Unlike public users, Daniel sometimes needs to view individual-level records, including donor names, donation amounts, dates, political entities, and postal codes.

Profile

Age	26
Education	Master's student in Political Science
Occupation	Political Science Graduate Research Assistant
Location	Toronto, Ontario
Tech literacy	High
Access level	Authenticated research user, login required
Data visibility	Individual-level donation records with privacy protections
Platform	Browser-based web app
Personality tags	Analytical, Detail-oriented, Responsible

Primary Task

Search and analyze individual-level donation records for academic research while respecting privacy and audit requirements.

User Story

As an authenticated research assistant, I want to access individual-level donation records through a secure login so that I can conduct detailed political finance research with appropriate privacy controls.

Core Needs

- Securely log in to the research tier.
- Search and filter individual-level donation records.
- Analyze donations by party, date, amount, postal code, and electoral district.
- Compare detailed records with aggregate map patterns.
- Understand which geographic boundary data is being used.

- Use individual-level data responsibly with privacy protections and audit logging.

Frustrations

- Current research workflows rely too much on scripts and programming knowledge.
- Individual-level records create privacy and access-control concerns.
- Postal code errors can affect the accuracy of geographic analysis.
- Electoral boundaries have changed over time, which makes comparisons harder.
- Public aggregate views may hide details needed for deeper research.
- It can be difficult to trust results without clear data source and boundary information.

Quote

“The public map helps me find patterns, but my research sometimes requires the underlying records.”

Success Criteria

Daniel succeeds if he can log in, search individual-level records, connect them to map-based patterns, and understand the data context well enough to support research work.

SECTION 4

Use Cases

Use Case 1: Explore CDMP Donation Map

Description

A public user opens the CDMP website and explores the donation data through an interactive map. The user selects a region to view aggregated donation statistics of a region without logging in or using any technical knowledge.

User Story

As a journalism student, Maya wants to explore political donation data on an interactive map so that she can view a donation summary for a specific region without any technical knowledge.

Actors

Primary actor: Public User

Supporting actors: None, user is interacting with the system directly.

Normal Flow

1. User opens the web application
2. System presents the donation map showing all of Canada
3. User selects a region to explore
4. System presents a donation summary for the selected region including total donations, party breakdown, and longitudinal trend information
5. User reviews the donation summary

Alternatives

A1.3 - User searches for a specific region

1. System presents donation summary for the searched region

Go to step 5

A2.5 - User returns to the national view of the map to explore another region

1. System presents national donation map showing all of Canada

Go to step 3

Exception

E1.2 - System fails to present the donation map when user opens the application

1. System displays an error message informing the user that the map is currently unavailable

End

Acceptance Criteria

1. Public user can access the map without logging in
2. System presents donation map when opening the web application
3. User can select any region to view donation summary
4. System presents total donations, party breakdown, and longitudinal trend information for selected region

Dependencies

Persona 1: Maya

Priority

High

Use Case 2: Filter Donation

Description

A public user uses filters to narrow down the donation data on the map by selecting specific filters such as the. The system updates the donation view based on the selected geographical level, political party, time period, per-capita metric, and socioeconomic filters allowing the user to explore specific patterns in the data.

User Story

As a civic transparency communicator, Amira needs to filter and view privacy-safe aggregated donation maps so that Amira can explain political finance patterns to her audience in a clear and non-technical way.

Actors

Primary actor: Public User

Supporting actors: None, user interacts with the system itself. There are no external systems.

Normal Flow

1. User requests to filter the data options
2. System presents the available filter options
3. User selects one or more filters of their choice
4. System presents the updated donation data based on the selected filters
5. User reviews the filtered data

Alternatives

A1.4 - User requests to view donation data per-capita

1. System presents donation data adjusted per person based on the selected filters

Go to step 5

A2.4 - User resets filters and select new ones

1. System presents the available filter options
2. User selects new filters of their choice

3. System presents the filtered donation data
Go to step 5

Exception

E1.4 - System presents insufficient donation data based on selected filters

1. System informs the user explaining why the selected region has limited donation data
2. System prompts the user to adjust their selected filters

End

E2.4 - System fails to present the filtered donation data

1. System informs the user that the filtered donation data is currently unavailable
2. System prompts the user to adjust their selected filters

End

Acceptance Criteria

1. Public user can filter the donation data without logging in
2. System presents the available filter options when user requests to filter the donation data
3. User can select one or more filters of their choice
4. User can filter donations by party for basic comparison analysis
5. System presents an insufficient data message for regions with limited donation data with selected filters
6. User can request per-capita view of the filtered donation data

Dependencies

Use Case 1: Explore Donation Map

Persona 2: Amira

Priority

High

Use Case 3: Access Individual Donation Data

Description

A user logs into an authenticated account within the CDMP platform and gets the individual donation records, such as donor name, exact contribution amount, date and postal code. Only users who are a part of the University can access this, and all the activity is logged.

User Story

As a graduate student doing a research role Daniel wants to be able to access individual donation records so Daniel can create a better statistical analysis on patterns within political finance.

Actors

Primary actor: Research User (authenticated)

Supporting actors: Authentication system (university identity provider)

Normal Flow

1. User navigates to the CDMP login page.
2. User enters university email and password.
3. System authenticates the user and verifies their research tier authorization.
4. System presents a privacy agreement describing how the data may be used.
5. User agrees to the privacy agreement.
6. System logs the agreement and access event into the activity log.
7. System provides access to a research dashboard containing individual donation records.
8. User specifies filter criteria of their choice.
9. System provides a paginated table of individual donation records matching the selected filters.
10. System logs the query to the activity log.
11. User reviews the records.

Alternatives

A1.9 - User exports the filtered records as a CSV file

1. User requests an export of the filtered records.
2. System generates a CSV file for the user.
3. System logs the export event to the activity log.

Go to step 11.

A2.8 - User modifies the filter selection

1. User changes the filter selection.
2. System generates updated donation records matching the filters.
3. System logs the new query to the activity log.

Go to step 11.

A3.5 - User declines the privacy agreement

1. User declines the privacy agreement.

2. System denies access to individual donation records.
3. System returns the user to the login page.

End

Exception

E1.3 - System fails to authenticate the user

1. System informs the user that their account is not authorized and advises them to contact the Policy Horizons Lab.
2. System logs the failed authentication attempt to the activity log.

End

Acceptance Criteria

1. People who are a part of the university can only access individual records
2. The system logs each and every login, query, and export to the activity log
3. The user can filter by party, year, and riding
4. The user can export the records as a CSV
5. The system denies access and logs if the user is not authorized
6. The user must agree to a privacy agreement before accessing individual donation records

Dependencies

Use Case 1: Explore Donation Map

Use Case 2: Filter Donation

Persona 2: Daniel Park

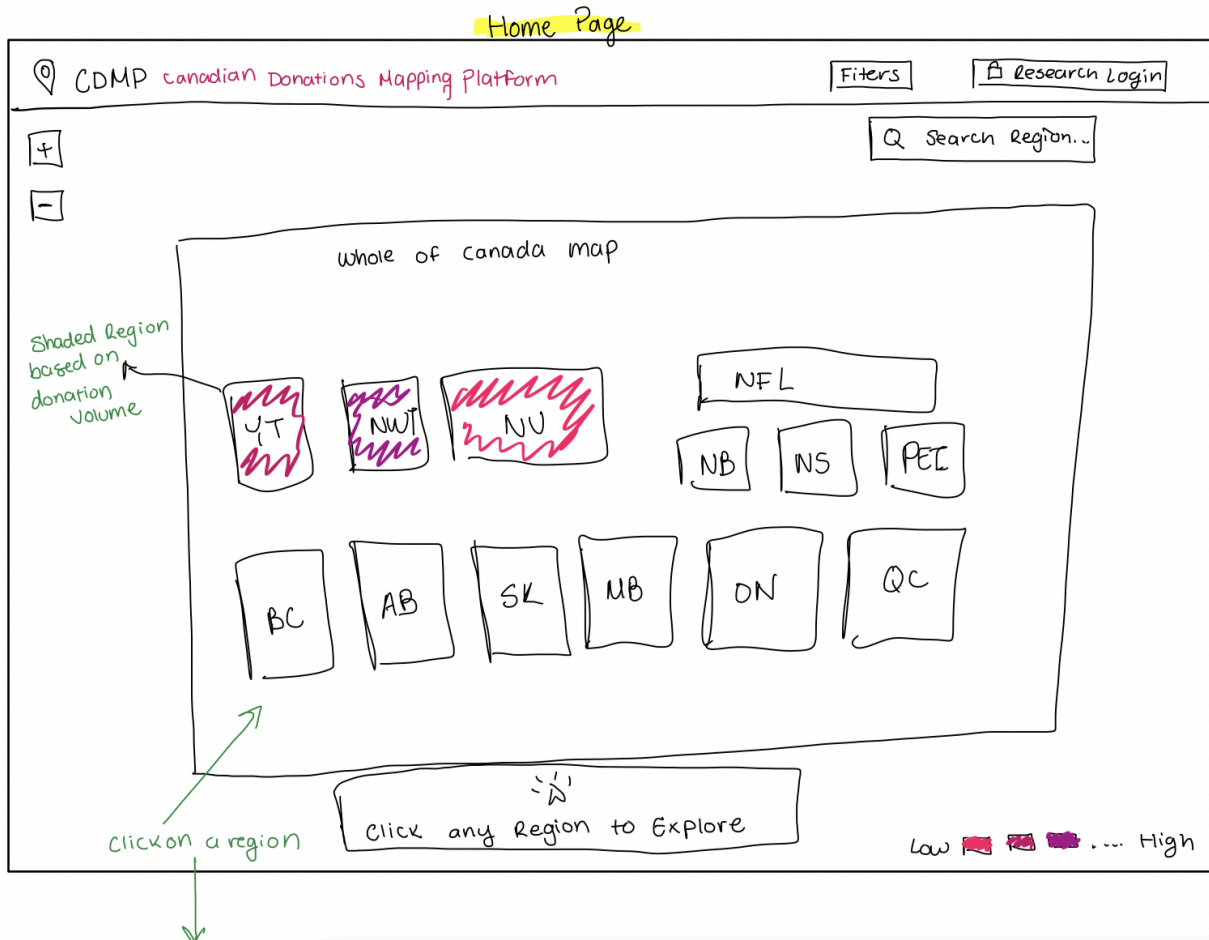
Priority

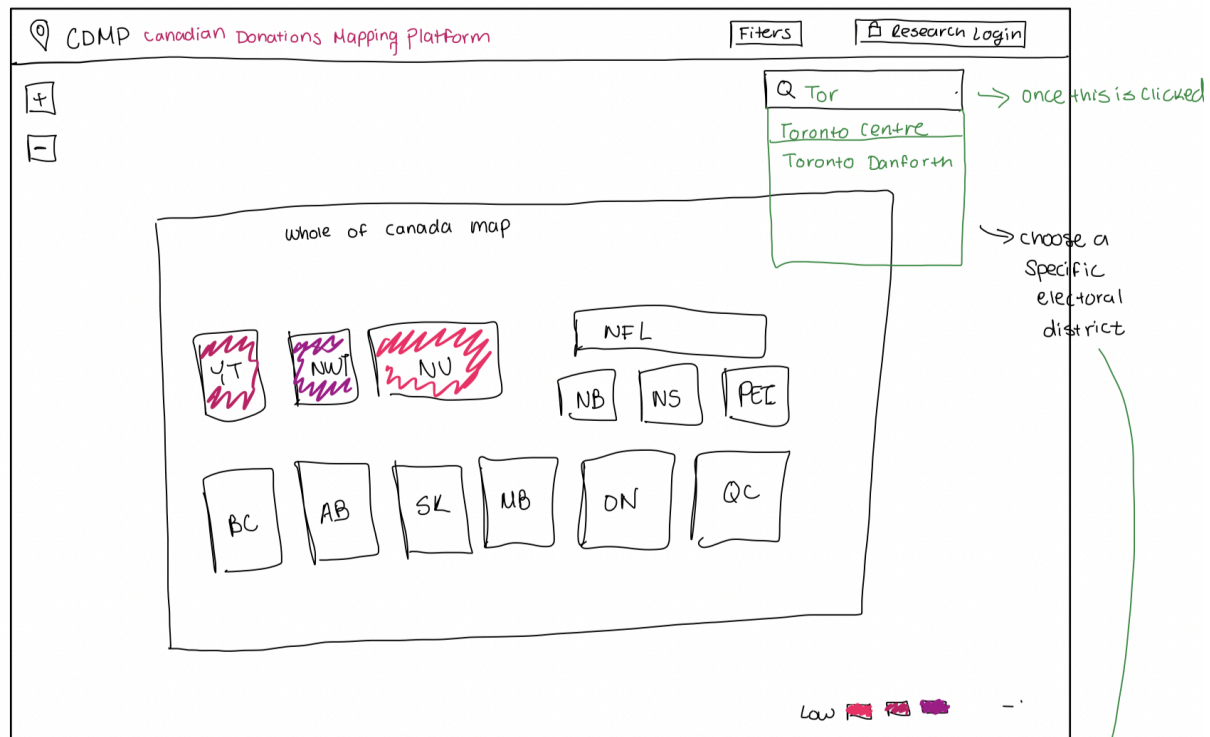
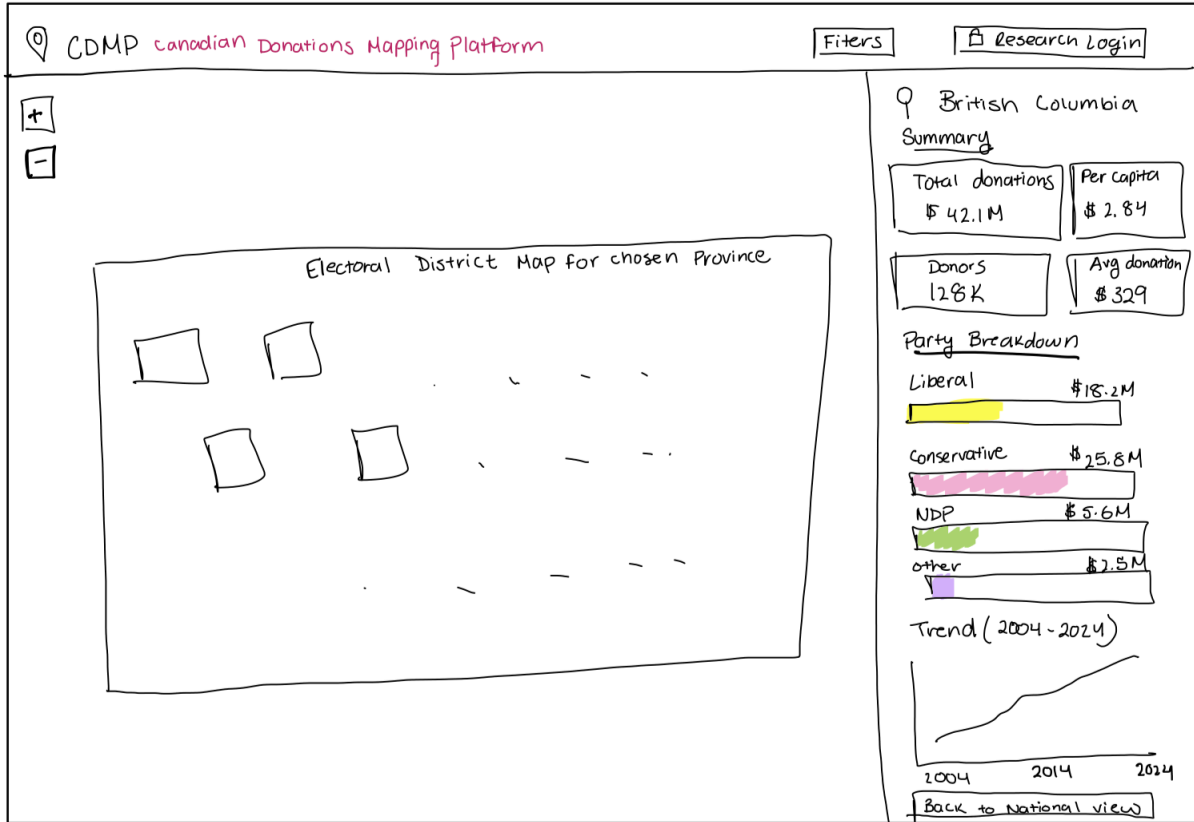
High

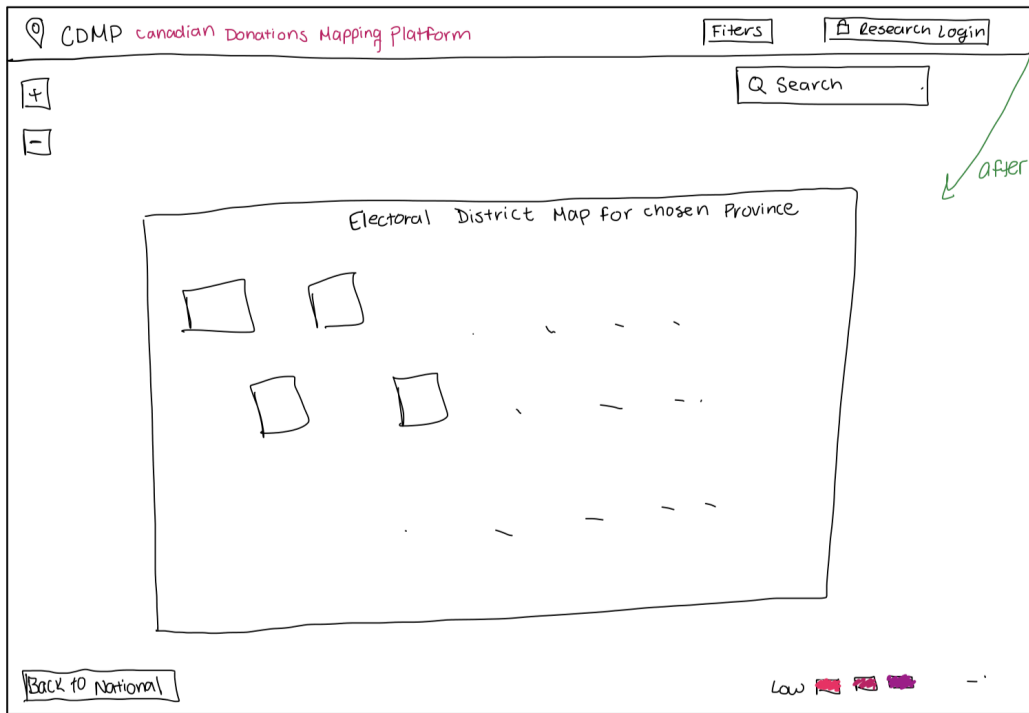
SECTION 5

UI Sketches

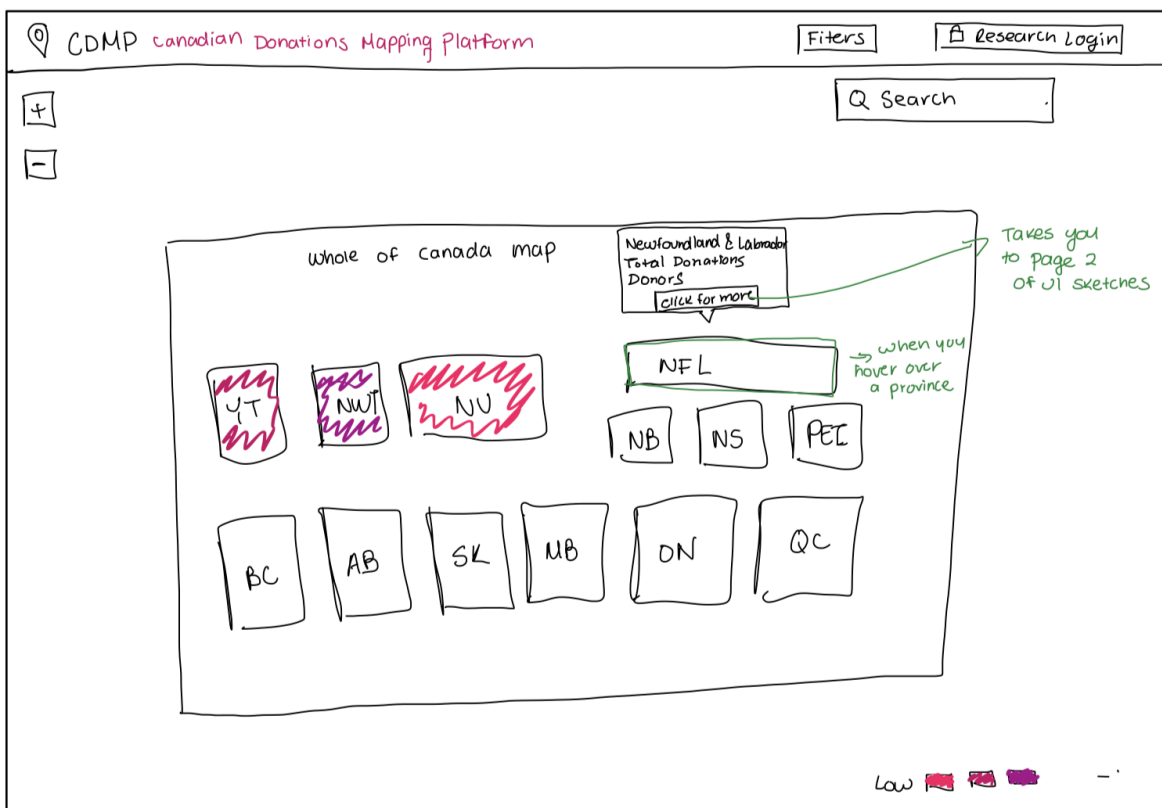
Use Case 1 - UI Sketches







after choosing district, jumps to this view



Takes you to page 2 of UI sketches

when you hover over a province

Use Case 2 - UI Sketches

Filters Page

CDMP

Research Login

Filters

Geography Level

Province default

↳ 0 ON

↳ 0 QC

↳ 0 Electoral District

↳ 0 ...

↳ 0 ...

Political Party

☐ All Parties

☐ Liberal

☐ Conservative

☐ NDP

Time Period

2004 +0 2024

Full range selected

Normalization

Total \$

Per capita

whole of canada map

YT

NWT

NU

NFL

NB

NS

PEI

BC

AB

SK

MB

ON

QC

Under Province a dropdown will pop-off

List changes depending on what province is chosen

CDMP

Research Login

Filters

Geography Level

Province default

ON

Electoral District

Political Party

All Parties

Liberal

Conservative

NDP

Time Period

2015

to

2023

Normalization

Total

Per capita

reset filters

Showing:

Ontario

Liberal

2015-2023

Per capita

Electoral District for Ontario

Another Exceptions Page

CDMP Canadian Donations Mapping Platform

Filters

Research Login

+

-

Limited Data Available

This region has relatively low donation activity due to its smaller population.

disclaimer also present for territories

Electoral District Map for chosen Province

Northwest Territories Summary

Total donations

\$42.1M

Per capita

\$2.84

Donors

128K

Avg donation

\$329

Party Breakdown

Liberal

\$18.2M

Conservative

\$25.8M

NDP

\$5.6M

Other

\$7.5M

Trend (2004-2024)

2004

2014

2024

Back to National view

Use Case 3 - UI Sketches

Screen 1: Sign up page

CMP Canadian Donorship Mapping Platform Login



Research Account Sign Up

Access to individual records is restricted to ~~only~~ approved university researchers

University email

Full name

University

Department/Faculty

Research Purpose

☐ I can confirm I am affiliated to this university

[Request access](#)

Screen 2: Research login page

CDMP Canadian Donations Mapping Platform

Filter Research Login

Research Login

Authorized by university researchers only

University email
Name@utoronto.ca

Password

Login

Screen 3: Research Dashboard (individual donations record)

CDMP Canadian Donations Mapping Platform

Filters Daniel Ark V. Research User

Research Dashboard

Search donors...

Donor	Amount (\$)	Party	Date	Postal code	Riding
John Smith	250.00	Liberal	2021-06-15	M5V 2T6	Toronto-
Sarah Ahmad	100.00	Liberal	2019-08-22	K1A 0A6	Ottawa centre
Michael Brown	300.00	Liberal	2019-09-10	V6B 1A1	Vancouver Ct
Emily chen	300.00	Liberal	2020-09-02	H2Y 1C6	Hocheleg
...	...	...	...	...	...

Filters

Geography Level
☐ Province
☐ Electoral
☐ FSA (Base area)

Political Party
☐ All Parties
☐ Liberal
☐ Conservative
☐ NDP
☐ other

Time Period
 2015
 to
 2024

Donation Amount (\$)
 Min
 to
 Max
 Apply Filters
 Export CSV

Research Activity
 Last Login
 May 26, 2025
 5:42 PM
 Recent Export
 May 26, 2025
 5:55 PM
 (1048 records)
 Activity Logged
 All login queries and exports are recorded.

Access Status
 Research Tier
 Active

Session 4: Export CSV

CDMP Canadian Donation Mapping Platform Filters Research User

Research Dashboard

Search donors...

Donor	Amount (\$)	Party	Date	Postal code	Riding
John Smith	250.00			M5V 2T6	Toronto-
Sarah Ahmad	100.00			K1A 0A6	Ottawa centre
Michael Brown	300.00			V6B 1A1	Vancouver C+
Emily Chen	300			H2Y 1C6	Montreal
...	...	...	...	...	...

Export Records
you are about to export the current filtered donation records.
this action will be logged
Do you want to continue?

Filters

Geography Level
☐ Province
☐ Electoral
☐ FSA (federal area)

Political Party
☐ All Parties
☐ Liberal
☐ Conservative
☐ NDP
☐ Other

Time Period
 2015
 to
 2024

Donation Amount (\$)
 Min
 to
 Max

Research Activity
 Last Login
 May 26, 2025
 5:42 PM
 Recent Export
 May 26, 2025
 5:38 PM
 (1248 records)
 Activity Logged
 All logins, queries and exports are recorded.

Access Status
 Research Tier
 Active

Screen 5: Authentication failure

CDMP Canadian Donation Mapping Platform Filters Login



Access Denied
 your account is not authorized to access the individual record.

Please contact the Policy Horizons Lab administrator